

Table 1: Closed-loop component ablation on 15 matched 60-second MemCam trajectories, $B = 32$. All settings use the same generation seed and reader.

Setting	FVD ↓	LPIPS ↓	PSNR ↑	SSIM ↑
Full (KEEPSAKE)	703.28	0.59708	11.739	0.30441
Without appearance	687.98	0.59689	11.726	0.30129
Without pose	716.11	0.59902	11.684	0.29905
Without closest-substitute term	697.32	0.59705	11.764	0.30308
Without degree term	720.57	0.59912	11.684	0.30048

Removal	Δ LPIPS [paired 95% CI]	Δ FVD [paired 95% CI]
Appearance	−0.00019 [−0.00505, +0.00417]	−15.30 [−69.77, +69.42]
Pose	+0.00194 [−0.00201, +0.00571]	+12.83 [−62.24, +79.04]
Closest-substitute term	−0.00004 [−0.00603, +0.00631]	−5.96 [−98.04, +46.29]
Degree term	+0.00204 [−0.00312, +0.00701]	+17.29 [−54.08, +84.91]

Δ = variant minus full; positive indicates worse LPIPS/FVD. Intervals resample whole trajectories: 5,000 draws for LPIPS, 2,000 for cohort FVD. All four primary LPIPS contrasts have Holm-adjusted $p = 1.000$. FVD intervals are descriptive. FVD is computed on pooled clips, not averaged per video. Scalar fidelity excludes the initial conditioning image.

Table 2: One-at-a-time parameter sensitivity on 15 fixed 60-second histories, $B = 32$. The evaluated quantity is the retained-bank oracle DINO-distance gap.

Setting	τ	β	λ	Gap ↓	Δ [paired 95% CI]	Jaccard
Default	3	0.5	0.25	0.03767	Reference	1.000
$\tau = 1$	1	0.5	0.25	0.03767	0.00000 [0.00000, 0.00000]	1.000
$\tau = 5$	5	0.5	0.25	0.03767	0.00000 [0.00000, 0.00000]	1.000
$\beta = 0$	3	0	0.25	0.03679	−0.00089 [−0.00703, +0.00641]	0.223
$\beta = 1$	3	1	0.25	0.03545	−0.00222 [−0.01108, +0.00499]	0.591
$\lambda = 0$	3	0.5	0	0.03970	+0.00202 [−0.00991, +0.01216]	0.371
$\lambda = 0.5$	3	0.5	0.5	0.03865	+0.00097 [−0.00146, +0.00348]	0.562

Δ = variant minus default; 5,000 paired trajectory-bootstrap draws. Jaccard measures retained-bank overlap with the default, not visual quality. All settings receive the same cached history. This CPU replay does not measure generated-video LPIPS/FVD or establish equivalence between settings.

Table 3: Post-hoc component diagnostic on the same 15 fixed histories, $B = 32$. No videos are generated in this CPU replay.

Setting	Gap ↓	Δ [paired 95% CI]	Jaccard
Full (KEEPSAKE)	0.03767	Reference	1.000
Without appearance	0.03385	−0.00382 [−0.01425, +0.00743]	0.298
Without pose	0.04248	+0.00481 [−0.00968, +0.01629]	0.236
Without closest-substitute term	0.03970	+0.00202 [−0.00991, +0.01216]	0.371
Without degree term	0.03964	+0.00197 [−0.00440, +0.00914]	0.226

Gap is retained-bank minus full-history oracle DINO distance; lower is better. Δ = variant minus full; 5,000 paired trajectory-bootstrap draws. Jaccard measures retained-bank overlap with the full method. This post-hoc diagnostic uses a shared history and a common DINO evaluation metric; it does not rerun the reader or generator.

Table 4: Recorded resources for KEEPSAKE-B32 with a CPU archive on NVIDIA H100 PCIe (PyTorch 2.5.1, CUDA build 12.4; 640×352, 50 denoising steps). Five starting views per round-trip protocol. Memory columns are maxima across the five runs; latencies are means of per-run averages. These are short-rollout diagnostics, not a matched comparison to unbounded retention or measurements of the 60/180-second cohort.

Protocol	RGB bank (MiB)	Host RSS (GiB) [†]	GPU alloc. (GiB)	GPU res. (GiB)	Retrieval (ms/query)	Desc. + update (ms/section)
90°, 153 frames	41.25	2.53	16.31	18.53	49.01	1684.58
360°, 609 frames	41.25	3.13	16.31	18.55	50.77	1523.65

[†]Maximum observed RSS at profiler checkpoints, not a continuously measured host peak. GPU columns use PyTorch rollout peak counters, not device-wide `nvidia-smi` usage. Descriptor extraction and policy update share one timer; they cannot be separated retrospectively. The RGB column excludes descriptors, metadata, output frames and model state. The maximum recorded retained-feature logical payload is 1.594 MiB; this does not account for Python containers or backing batches retained through array views. Retrieval excludes the initial section’s no-read fallback and includes no context encoding.